

PRAKHAR

Senior Software Engineer | Full Stack | Node.js | React | PostgreSQL
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Profiles

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Summary

Senior Software Engineer with **4+ years of experience** building enterprise SaaS, ERP, and Industrial IoT applications across the full software development lifecycle. Hands-on experience with Node.js, Express.js, React, JavaScript, and PostgreSQL, with ownership spanning frontend development, backend services, REST APIs, database design, integrations, deployment, and production support. Built and supported a **12+ module ERP platform with 300+ APIs** and production applications used across multiple customer organizations. Experienced in real-time industrial applications integrating PLCs and automation systems through MQTT, Modbus, and TCP/IP. Promoted from Software Development Engineer to **Software Manager**, combining hands-on engineering with technical leadership and end-to-end delivery.

Experience

YM Automation Pvt Ltd Jul 2022 - Present	Coimbatore, Tamil Nadu
Software Manager	Oct 2024 - Present
- Led a cross-functional software team across backend, frontend, mobile, and UI/UX while remaining hands-on with application architecture, development, debugging, and production support. - Designed and developed full-stack architectures for enterprise SaaS, ERP, and Industrial IoT applications, covering frontend interfaces, REST APIs, business logic, data models, integrations, and deployment. - Led 7-8 projects across ~10 customer companies, including multi-site industrial deployments and 24x7 IoT systems integrating PLCs through MQTT, Modbus TCP/RTU, and TCP/IP. - Managed end-to-end technical delivery including customer requirements, estimation, proposals, architecture, development planning, deployment, troubleshooting, and production support. - Conducted code reviews and mentoring, allocated technical tasks, tracked delivery, and resolved technical blockers across frontend and backend development.	
Software Development Engineer	Jul 2022 - Oct 2024
- Independently designed, developed, deployed, and supported 12 production software projects, contributing across frontend, backend, database, integrations, and production environments. - Developed full-stack web applications using React.js, JavaScript, Node.js, Express.js, and PostgreSQL, building responsive interfaces, REST APIs, business logic, authentication, RBAC, and application workflows. - Designed PostgreSQL schemas and implemented SQL queries, relationships, indexing, transactions, JSONB, and database optimization for production applications. - Developed real-time integrations with PLCs and industrial systems using MQTT, Modbus TCP/RTU, TCP/IP, TCP sockets, and polling, with typical polling intervals of ~1 second depending on requirements. - Built and supported 24x7 Industrial IoT applications for machine monitoring, production, energy, alarms, events, and analytics, integrating up to ~65 machines/devices.	

Projects

Enterprise ERP & SaaS Platform
- Designed and developed a 12+ module ERP/SaaS platform covering HRMS, projects, attendance, payroll, inventory, purchase, sales, CRM, and other enterprise operations. - Built full-stack functionality using React.js, Node.js, Express.js, and PostgreSQL, developing reusable UI components, REST APIs, business workflows, authentication, RBAC, validation, and data persistence. - Developed 300+ REST APIs and designed a PostgreSQL data model with 100+ tables, implementing transactions, JSONB, indexing, and reusable application patterns. - Built the platform to support 100+ companies, with 4 companies currently onboarded and approximately 300-400 employees represented in the system.

Production Tracking & Workflow System

- Developed a full-stack production tracking system supporting **12-stage production workflows with branching routes** based on operational requirements.
- Built frontend workflows and backend APIs to manage **stage transitions, routing, production status, quantities, and workflow progress**.
- Implemented application logic for tracking production movement across different workflow paths and providing visibility into ongoing operations.

Industrial IoT Monitoring Platform

- Developed full-stack production applications for **24×7 real-time monitoring** of industrial machinery and automation systems.
- Built React-based monitoring interfaces and Node.js backend services for machine status, production data, alarms, events, and historical analytics.
- Integrated PLCs and machines through **MQTT, Modbus TCP/RTU, TCP/IP, and TCP sockets**, implementing polling, data acquisition, processing, and persistence with typical ~1-second polling intervals.
- Designed PostgreSQL data models and application workflows supporting systems with up to **~65 machines/devices**.

Production & Energy Monitoring System

- Developed full-stack functionality for real-time and historical **production, energy, machine, alarm, and event monitoring**.
- Built REST APIs, database workflows, and frontend monitoring views for operational dashboards and analytics.
- Optimized SQL queries and database workflows using indexing, transactions, and efficient data-access patterns for production monitoring and reporting.

Skills

Frontend

React.js, HTML5, CSS3, Axios, State Management, Responsive UI, Javascript

Backend

Node.js, Express.js, JavaScript, REST APIs, WebSockets, Authentication, RBAC, API Development, Business Logic

Database

PostgreSQL, SQL, Database Design, Query Optimization, Indexing, Transactions, JSONB, MongoDB

Cloud & Deployment

Microsoft Azure, AWS EC2, Linux, Nginx, PM2, GitHub Actions, SSL, DNS

Industrial IoT

MQTT, Modbus TCP/RTU, TCP/IP, TCP Sockets, PLC Integration, Real-time Data Processing

Engineering

Full-Stack Development, System Architecture, Database Architecture, API Development, Performance Optimization, Authentication, RBAC, Code Review, Debugging

Education

Siddaganga Institute of Technology, Tumakuru

Computer Science and Engineering

Bachelor of Engineering • 6.62

Tumakuru, Karnataka • 2011 - 2018